

BookBot Bin Logic Challenge

Retrieve the right book by using bin addresses, not a conveyor race.

Win condition: Most correct retrievals with the fewest legal moves and a clear algorithm defense.

THE SCIENCE YOU ARE USING

Automated storage and retrieval. Temple's Charles Library BookBot stores books in bins identified by an address, not by subject on a shelf. A crane fetches the bin by its address on request. The same logic runs warehouses: store anywhere, remember the address, retrieve on demand.

Routing and search. Good addressing and smart routing minimize legal moves. You simulate storage and retrieval with order cards, bin addresses, and routing rules, then defend your algorithm.

HOW TO RUN IT

- 01 Set up the bins.** Use one shared tabletop judging mat with rows A-D and columns 1-6. Mark DEPOT immediately left of A1. Teams may use separate copies to plan, but they run the shared mat one at a time.
- 02 Read an order.** Draw an order card listing the items to retrieve and their addresses.
- 03 Plan the route.** Start at DEPOT, order the retrievals, and return to DEPOT. One legal move goes to an orthogonally adjacent bin (never diagonal) or between DEPOT and A1. Route cost is the number of legal moves.
- 04 Run the retrieval.** Move one legal step at a time, retrieve each requested address, and return to DEPOT. Record legal moves before time; a missed address or illegal jump makes the run incomplete.
- 05 Defend the algorithm.** Explain the rule you used to choose the order and why it is efficient.

Safety: Keep clear of actual library operations. Tabletop version preferred after tour. Allergy: None.

MATERIALS

Bin cards	set
Address labels	shared
Order deck	per team
Planning route mats and shared judging mat	one planning copy per team; one shared judging mat
Timers	per team
Grabber tool	optional

YOUR PLAN AND DATA

Record the card addresses, route order, legal move count, and card optimum for the shared-mat run.

Order card and required addresses: _____

Planned route from DEPOT back to DEPOT: _____

Team legal moves and card minimum: _____

Efficiency score = $\max(0, 10 - (\text{team moves} - \text{card minimum}))$: _____

DEFEND YOUR ROUTE

What rule did your team use to choose the retrieval order? _____

Which route segment could be shortened while keeping every move legal? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Legal complete route	25
Routing or algorithm explanation	20
Efficiency versus card optimum	10
Teamwork	10
Total	100